

ASCEND: An AI driven 360° career intelligence & personal growth companion

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Abstract: The move from academics to a career in industry typically presents numerous obstacles to students resulting in poor preparation and absence of structured feedback. Existing services generally focus on providing isolated solutions like coding practice and fail to provide overall student's career readiness status. Therefore, this paper presents ASCEND , an innovative Artificial Intelligence (AI) based career readiness platform that simulates the complete recruitment process from beginning to end. ASCEND evaluates resume quality, technical skills, communication skills, interview performance across multiple modules of employability. The feedback and performance scoring of ASCEND is done by three technologies: Natural Language Processing(NLP), Large Language Model(LLM)-based Semantic Analysis, and Rule-Based evaluation techniques. These three technologies provide feedback and performance score. ASCEND provides personalized career improvements roadmaps and recommendations based on skill gaps. Results from evaluations indicate that ASCEND provides users with accurate and reliable assessment feedback and demonstrates the ability of ASCEND system to provide users with real-time feedback. By combining technical , behavioral , communication within single system, ASCEND provides an innovative approach to closing the gap between academic preparation and industry expectations for students.

Keywords: Resume ATS analysis, AI mock interview, Technical evaluation, Communication evaluation, Career Recommendation System , Natural Language Processing.

1. INTRODUCTION

In today's job market being highly educated is no longer enough to get hired. Employers want to see their new employees have more technical abilities , they want to know how well candidates can communicate with others, how well they solve problems. Unfortunately, many students don't have these qualities when they are moving for interviewing for jobs.

One of the biggest issue faced by students in preparing for their future career is that there is no single resource available to prepare for their careers as a whole. Students often use multiple platforms for preparing to various career paths , including coding practices, creating resumes, and preparing for interviews. While all these platforms do not provide a complete picture of an individuals employability. Therefore, students receive incomplete feedback about what they are doing well and what they need to work on, and therefore many students have no idea what they still need to do in order to become employable.

As such, with the emergence of numerous software programs that utilize Artificial intelligence(AI) and Natural language Preprocessing(NLP) to provide consistent preparing candidates for interviews. This research paper will discuss the creation of ASCEND, an AI-driven Career Readiness Platform, which takes full advantage of these new technologies by developing a real-time hiring process used in the industry.

1.1 Research Motivation

ASCEND is founded on the need for a better link between academic preparation and recruitment criteria established by employers. Although students will develop theoretical knowledge, many will be unprepared for the recruitment phase due to limited access to resources to improve their readiness for employment.

Historically, most resources for student preparation focused on evaluating one part of the employment process. Coding services may evaluate programming skills but do not assess communication skills. Resume-building services primarily focus on format rather than industry relevance or ats compatibility. Mock interviews, usually

happen in an informal setting. All of this together approach results in minimal feedback for students about what they are doing well or what skills they need to develop.

As the recruitment process continues to change and include the automated evaluation systems for screening candidates and AI-based hiring evaluations, preparing candidates for hiring process is very essential. By utilizing the advanced technologies like artificial intelligence and large language models to simulate these processes in an intelligent manner: evaluating answers in context, identifying performance gaps and providing feedbacks.

Hence the motivation of this system is to build an AI powered integrated platform that assess candidates while guiding them towards their career development.

1.2 Key Contributions

This research paper presents new and significant findings that have led to several contributions to the field of AI-based career readiness:

- **Integrated Career Readiness Software:**
ASCEND combines resume evaluation, technical aptitude, communication skills evaluation, AI interviews into one software by integrating multiple tools into one system.
- **Semantic Analysis of Resume and Interviews:**
Using NLP and LLM will allow recruiters to assess resume submission and performance of candidates in the interviews.
- **Adaptive Assessment System:**
ASCEND has a system that modifies the difficulty level of the questions asked during the job interview.
- **Identification of skills needed by individual applicants:**
ASCEND identifies specific skills that an individual needs to develop or improve upon in order to meet employer expectations and provide step-by-step instructions for improving those skills over time.

2. LITERATURE SURVEY

AI-Based Career Guidance Systems

Recent research indicates the Somesh with AI-based career guidance system called Sankalp analyzes emotions using recommendations to help students for career related decisions. Zhang Yanan further developed a conceptual framework that leverages semantics technology to provide a method for mapping student recommendation to various job positions using a model. Most systems provide guidance on what jobs to pursue but typically do not offer comprehensive assessments of a student's readiness for recruitment.

AI-Based Mock Interview Systems Evaluating Communication Skills

Several research efforts have explored the use of technology-based training such as those that simulate mock interviews. Kansada Mongkolijaturong in 2025 with a meta human-based mock interviewing tool that offers student's performance feedback. Another study by Edward Londner in 2023 offered a way to evaluate technical communication skills based on assessments. All of these platforms have been shown to improve student's confidence but do not work as a complete career readiness model.

AI-Learning and Skill Prediction Systems

AI has been utilizing for providing personalized learning environments. Pardamean in 2022 proposed a system that can predict the student's preferred style of learning. Although it is useful for academics but they do not support recruitment preparation.

Research Gap Analysis

Reviewing the available literature on this topic, it is evident that majority of the available solutions focus on one or more isolated modules of career development. They do not present any integrated systems that assess all of the modules such as assessing resume, technical skills, communication skills, job interview performance, roadmap,

and opportunity. As a result, the proposed ASCEND is an artificial intelligence platform that will bring together all of these into one place and provide overall readiness industry score and personalized individual improvement plan

3. PROBLEM STATEMENT

Transitioning from academic education to workforce is a challenging experience for students. Academic institutions provide students with theoretical knowledge and technical basis for their careers. Unfortunately, many students find themselves unprepared for what employers expect during recruitment process. Currently, students are utilizing various resources to prepare themselves for their careers, such as coding websites, to build resume and for interview preparation. These work independently and students receive little feedback on their performance.

Another challenge in preparing students for a successful transition is that more companies are using automated recruitment technologies like Application tracking System(ATS) and AI-based screening technologies , to assess applications. As a result, many students do not understand how this ATS and AI technology assess their resumes and responses even though they have the skills needed to perform in the role that they applied for.

Thus, an integrated , smart platform is much needed to create a simulation for recruiting workflows, measuring a number of employability dimensions, and providing organized feedback. The proposed system-ASCEND-offers an AI-driven employer readiness platform that brings together resume analysis, technical assessment, communication evaluation, mock interview simulation, opportunities, roadmap in one ecosystem for a comprehensive evaluation of readiness for the workforce.

4. PROPOSED METHODOLOGY

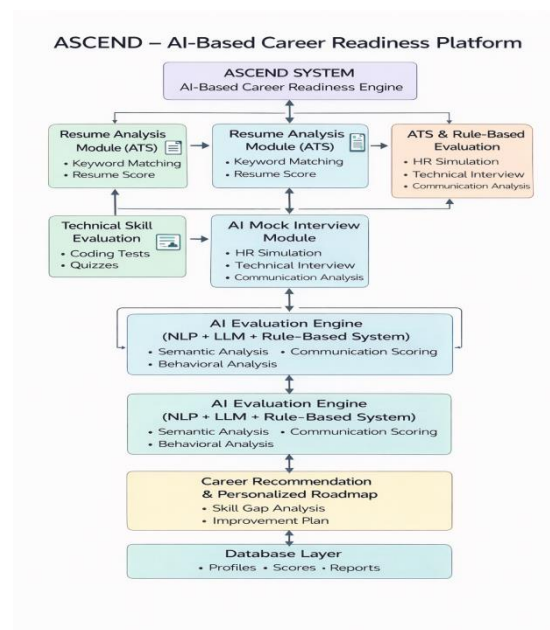


Fig.1: Flow chart

The ASCEND system will first collect input from user and then apply the input through the series of evaluation phases. Each phase consists of a unique algorithm used to calculate performance metrics of the user submitting the resume. Finally , the performance metrics calculated from all assessments are combined to generate a total industry readiness score.

Resume ATS Analysis: the resume application system(ATS) evaluation is based on approach of evaluating resume with ats tracking algorithm. Initially, the user uploads his/her resume to the ASCEND system , this system parses the resume to extract the text. Once the text has been extracted it tokenizes it process with relevant words based on keywords.

- K_m = Matched keyword set
- K_t = Total role keywords

Similarity between resume and job role is computed using semantic matching:

$$ATS_{score} = \frac{K_m}{K_t} * 100$$

Higher similarity indicates stronger ATS compatibility.

LLM-Based Semantic Evaluation: ASCEND utilizes semantic analysis of the candidate's responses, using Large Language Models, for students communication assessments and AI-mock interviews and Gemini LLM's prompt-based evaluation will process text and speech responses.

Semantic similarity is computed using keyword relevance scoring:

- W_c = candidate response words
- W_e = Expected answer words

$$Score_{semantic} = \frac{|W_c \cap W_e|}{|W_c \cup W_e|}$$

This hybrid evaluation combines statistical overlap with LLM contextual interpretation.

Technical Skill Assessment: Technical proficiency is evaluated using Automated Test Case Validation Algorithm. Candidate code submissions are compiled and executed against predefined test datasets.

- T_p = passed test cases
- T_t = Total test cases

$$Score_{tech} = \frac{T_p}{T_t} * 100$$

This deterministic approach ensures objective coding evaluation.

Speech linguistic Feature Analysis: Communication abilities will be analyzed against a Speech Linguistics Algorithm. Speech will have a computed transcription from audio to text, before being evaluated through Natural Language Processing techniques.

Key indicators of linguistic ability will include:

- Number of filler words used
- Connection density
- Rate of speech
- Fluency index

Scores will be normalized to assess overall proficiency in communication.

Opportunity Recommendation: The opportunity module uses a Skill Matching Recommendation Algorithm to map candidate competencies with job role requirements.

- S_u = User skill set
- S_j = Job requirement skill set

$$Score_{match} = |S_u \cap S_j| / |S_u \cup S_j|$$

This algorithm recommends relevant job roles , internships , scholarships and career opportunities aligned with candidate readiness.

Learning Roadmap generation: the skill Gap Analysis Algorithm is utilized by the roadmap module in order to create a customized development plan.

- S_{target} = Required industry skill level
- S_{current} = Candidate skill level

$$\text{Gap} = S_{\text{target}} - S_{\text{current}}$$

The system assigns established course and certificate level recommendations based on gap size.

Industry Readiness Scoring: The final score is calculated via weighted aggregation algorithm. Aggregating all the module's output.

$$\text{Score}_{\text{final}} = (W_{\text{res}} \cdot S_{\text{ats}}) + (W_{\text{tech}} \cdot S_{\text{code}}) + (W_{\text{comm}} \cdot S_{\text{speak}}) + (W_{\text{int}} \cdot S_{\text{mock}})$$

This score is determined as the candidates industry readiness level.

5. EVALUATION METRICS

The performance of the proposed ASCEND system is evaluated using standard classification metrics.

$$\text{Accuracy} = \frac{TP+TN}{TP+TN+FP+FN}$$

$$\text{Precision} = \frac{TP}{TP+FP}$$

$$\text{Recall} = \frac{TP}{TP+FN}$$

$$\text{F1-Score} = \frac{2TP}{2TP+FP+FN}$$

Roc-Auc:

$$TPR = \frac{TP}{TP + FN}$$

$$FPR = \frac{FP}{FP + TN}$$

$$\text{AUC} = \int_0^1 TPR(FPR) d(FPR)$$

To evaluate the effectiveness of the ASCEND system, standard classification-based performance metrics are adapted to measure the correctness and reliability of assessments outputs.

6. EXPERIMENT ANALYSIS

A number of employment assessment module-related performance tests will run on ASCEND career readiness platform, to determine its overall performance. ASCEND has been tested with candidate data sets, including resumes, technical tests, communication and interview interactions.

Data used to evaluate through the ASCEND system included:

- Candidate resumes
- Technical and aptitude test results
- Voices communication recordings and interviews
- Transcripts of interview answers

The career Readiness Dashboard provides a visual summary of the scores for each module and a total score for career readiness system. It shows student's current status of completion , level and progress towards achieving their goals through roadmap.



Fig.2:Dashboard interface showing overall modules evaluation status

The AI Mock Interview module simulates an interview in real-time through a Long Term Memory Model(LTMM) powered by Artificial Intelligence conversational evaluation of the responses submitted by candidate. The mock interview creates and records an audio-visual recording of the candidate's answers which will be converted to text through transcription and examined for their semantic relevance and clarity.

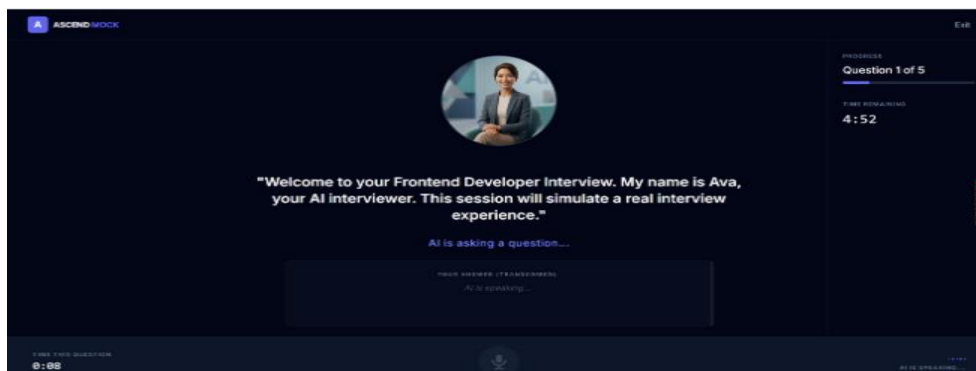


Fig.3: This image shows the mock interview simulation interface with candidate's responses to AI questions.

The resume ATS can identify if there is enough structure in a resume. By using a structured resume to rank a candidate could find them quickly due to their ATS compliance. If any sections of the resume are missed ATS will provide feedback to the candidate on what is missing and how to correct it.

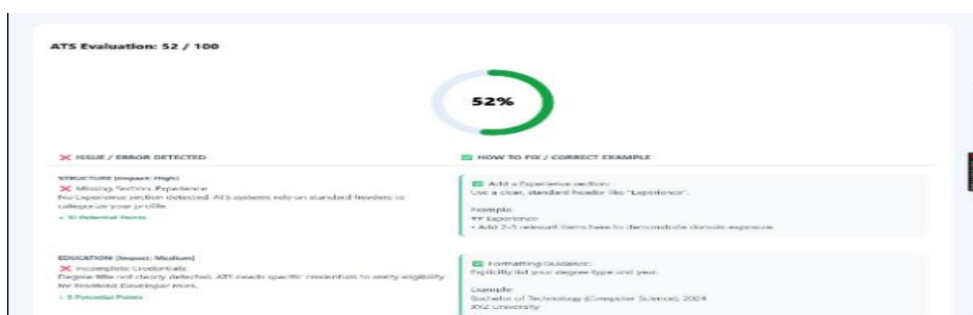


Fig.4: This image shows the evaluation report by the ATS .

The performance of the candidate in ASCEND was calculated using accuracy, precision, recall, f1-score, roc-auc metrics. Logistic Regression was rated with the highest percentage of 91%, with precision 79.41% , recall 93.10% and f1-score 85.71% , it indicates that the strong reliability. The Random Forest model was rated with 88% accuracy, 80.00% f1-score and roc- auc with 0.9573. After all, the results indicates that Logistic Regression is the better and stabilized engagement prediction within ASCEND system.

Model	Accuracy	Precision	Recall	F1-Score	ROC-AUC
Logistic Regression	0.91	0.7941	0.9310	0.8571	0.9825
Random Forest	0.88	0.7742	0.8276	0.8000	0.9573

Table.1: Comparative Model performance.

7. CONCLUSION

ASEND is a platform designed to help students move from education to employment by providing an AI-assisted system designed to provide them with the tools to make a successful transition from school to work. Most current applications only address one part of the career preparation process. For example, they may check resumes or provide a coding environment where users can practice writing code. Because ASCEND integrates all aspects of preparation together into one seamless system, whether it is providing students with simulated interviews using AI, ASCEND will provide students with an authentic experience of what it feels like to navigate the entire recruitment process.

Data collected from students who have used ASCEND experimentally demonstrates that ASCEND provides students with timely and consistent feedback in addition to providing students with reliable feedback, which has resulted in increased confidence in their ability to navigate through the recruitment process and to be more aware of their career readiness. ASCEND employs an intelligent integrated to connect students technical skills, behaviour skills and communication skills to provide students with a concrete, logical and scalable solution to the discrepancy what students learn in a classroom and what employers are looking for in graduates when they complete school.

To summarize, ASCEND is an opportunity to continuously develop smarter, better, more effective systems for supporting the preparation of students as they enter the recruitment process, enabling them to more confidently and better prepared for job opportunities.

8. FUTURE WORK

While ASCEND currently offers a robust career preparation assessment structure, there are many avenues through which it can be developed beyond its current scope in future versions.

One strong potential direction is adding forms of AI-based tools to enhance the evaluation of behaviour and emotional throughout mock-interviews through;

- Facial expressions
- Tone of voice
- Body Language

Incorporating these assessments will provide a more comprehensive evaluation of each users' behaviour and emotional intelligence while delivering a more realistic simulation of the recruitment process.

Existing ASCEND modules can also be expanded into industry- specific domains which will enhance the relevance and accuracy of the applicable assessment criteria for software Engineering ,Data Science , Management, Core Engineering and Government jobs.

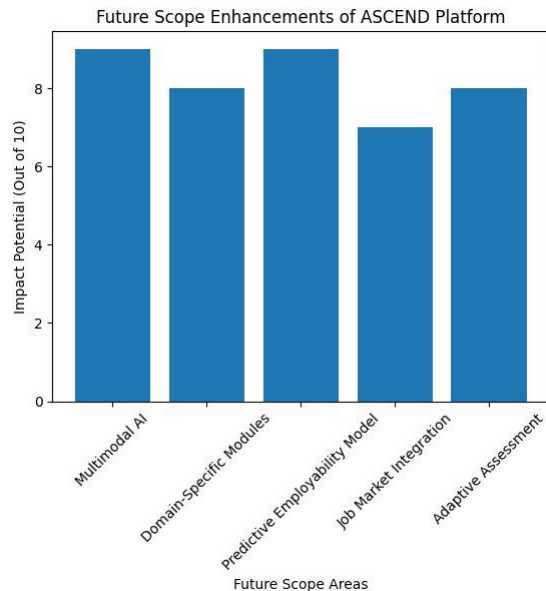


Fig.6: Future scope of ASECND

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